

Cross-modal scenario generation for stock price forecasting using Wasserstein GAN and GCN

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ARTICLE INFO

Keywords:

Stock price forecasting
Scenario generation
Cross-modal information fusion
Stock correlation
Deep learning

ABSTRACT

The forecasting of stock price has always been a difficult problem, as the various inputs like company performance, technical innovation, political factors are intricate and often assessed with uncertainty. While most of the existing studies belong to deterministic forecast that focused on either the trend or exact value of future stock price, the forecasted results contain little uncertain information, thus bringing certain risks to investors. In this study, an effective scenario generation method based on wasserstein generative adversarial network (WGAN) is proposed for stock price forecasting, which can characterize the temporally and spatially correlations among different stocks, so as to help investors make proper investment strategies in complicated market environments. To the best of our knowledge, the scenario generation of stock price has been rarely studied by existing publications. Specifically, considering the inherent correlation among stocks, graph convolutional network (GCN) is used to capture the features of related stocks, whereafter the features of target stocks and related stocks are combined. In terms of feature fusion, this paper further uses a pre-trained model to obtain the key information of news headlines as text features, and fully integrates them with historical data through the attention mechanism, thus improving the overall performance of scenario generation. In summary, the proposed model can generate promising scenarios for future stock prices over time based on cross-modal information like historical data and news headlines. Numerical experiments show that our method outperforms other baseline methods in terms of root-mean-square error and average absolute percentage error (the average of several scenarios). For each part of the method, stock correlation is the most helpful for the results, followed by cross-modal information, with scenario generation contributing less to the results. Besides, experiments on a real-life portfolio selection problem also demonstrate that our method brings the highest returns, which proves its effectiveness in practical application.

1. Introduction

Stock price forecasting [1] is an important research direction in the financial field, which is to use various scientific methods to provide the future price or trend of the stock. Accurate forecasting results can help investors better understand market trends [2] and formulate corresponding investment strategies. The importance of stock price forecasting is also reflected in risk management [3]. Different investors have different risk tolerance and investment goals. By predicting the stock price, investors can better estimate the risk of investment and take corresponding risk management measures, such as controlling positions as well as setting stop loss points.

Deterministic forecast refers to the generation of unique prediction results under specific conditions, whereas uncertainty forecast can

yield multiple potential prediction outcomes. The majority of existing research in this field has focused on deterministic forecast, which primarily encompasses two categories: point forecast and trend forecast. Point forecast aims to provide an exact value for the stock price at a future point in time, based on historical data from a particular period. Conversely, trend forecast strives to offer insights into the overall direction and changes in future price trends. For example, the researchers used support vector machines (SVM) [4], empirical mode decomposition [5] and artificial neural network [6] to achieve point forecast of stock prices, as well as using a hidden Markov model [7] to achieve trend forecast of stock time series data.

Although deterministic forecast provides useful information for decision-making, there is an obvious disadvantage: the stock market is

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<https://doi.org/10.1016/j.asoc.2024.112342>

Received 15 May 2024; Received in revised form 11 October 2024; Accepted 12 October 2024

Available online 19 October 2024

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affected by a variety of factors, including politics, economy, technology, etc., which are often difficult to quantify and capture. When using a deterministic forecast model, the predicted value may significantly diverge from the actual value, leading to lower-than-anticipated investment returns or risks. Additionally, deterministic forecast methods rely heavily on the completeness and accuracy of historical data. When data is incomplete or contains noise, the forecast results may be severely affected.

Thus, forecast with uncertain knowledge has attracted great attentions in recent years, which includes interval forecast, probabilistic forecast and scenario generation. Interval forecast [8] is a method that provides the upper and lower bounds of uncertain parameters, its purpose is to reflect the possible situations so as to provide quantitative information. Probabilistic forecast [9] is a statistics-based method for predicting future events or outcomes expressed as one or more probability distributions [10], each representing the likelihood that the corresponding event or outcome will occur.

In contrast, scenario generation [11] is more intuitive, which describes uncertainties in a productive way. A set of multiple scenarios represents the uncertainty of the stock price in the future, while a single time series represents the trajectory of the stock price in the future. The generated scenario can directly serve stochastic optimization [12] and effectively avoid risk as well as aid in decision-making [13]. With these advantages in mind, significant achievements have been made on scenario generation in some other research fields, such as renewable scenario generation [14] and aerospace system engineering optimization [15].

Theoretically, scenario generation is also promising for stock price forecasting: using multiple scenarios in a set, scenario generation can provide actual and diversified knowledge about stock price uncertainties, to give a fuller picture of the complexity of the market. These allow investors to make more accurate decisions and assess potential risk factors. By considering multiple scenarios, investors can cope with market volatility [16] and uncertainty, improving the return of their investments. Therefore, it is meaningful to study the scenario generation of future stock prices under complicated market environment.

Besides, for a long time, the forecasting of stock prices relied only on historical data [17]. The development of natural language processing [18] has led to the use of unstructured text to enhance the model's forecasting capabilities. And researchers have proved that cross-modal [19] information can be jointly used to advance the forecasting performance [20]. As a result, cross-modal information like structured historical data and unstructured market news is also considered to effectively improve the quality of scenario generation in this paper.

Furthermore, there are positive or negative correlations among different stocks, and some highly correlated stocks may have concordant or converse price movements. By taking advantage of this correlation, investors can gain more through arbitrage or hedging strategies. Researchers are able to use correlation coefficients to determine the relationship between returns and volatility, so as to build better investment strategies [21]. Therefore, it also makes sense to consider correlations among different stocks in this research.

To sum up, the main research contents of this paper include: in order to achieve effective scenario generation, graph convolution network (GCN) is used to capture the correlation among multiple stocks, and cross-modal information including structured historical data as well as unstructured market news is considered. Then, Transformer decoder is used for forecasting. Finally, an improved version of generative adversarial network (GAN), wasserstein GAN (WGAN) is used for scenario generation. The validity and practicability of the proposed method are verified by simulated investment experiments.

The main contributions of this paper include: (1) A scenario generation method of stock price is proposed innovatively in order to deal with the complicated uncertainty of the stock market, which has not been investigated by existing studies. (2) Cross-modal knowledge

including historical data and market news is used to improve the effect of the scenario generation. In addition, considering the correlation among stocks, news and stock price characteristics of related stocks are captured. And feature fusion is carried out between cross-modal features by using the attention mechanism, to fully learn the inner characteristics. (3) The effectiveness of the proposed method is verified via not only statistic analysis on the scenario generation itself, but also experiment on forex markets as well as investment performance on a real life portfolio selection problem.

The remaining sections of this paper are arranged as follows. Section 2 introduces the research status of scenario generation, cross-modal stock price forecasting and stock correlation analysis in recent years. Section 3 presents the concept and visualization of stock price scenario generation, then introduces some basic knowledge. Section 4 explains the forecasting method proposed in this paper. Section 5 shows the experimental results and proves the effectiveness of the research. Finally, Section 6 is the summary and discussion of the paper as well as the prospect of the future.

2. Literature review

With the development of artificial intelligence, the financial sector has also been greatly affected. Ahmadi-rad [22] dissected the multifaceted role of artificial intelligence in the contemporary business ecosystem. Ahmadi-rad [23] delved into the heart of the changes of the banking and investment sectors, examining what challenges they may encounter moving forward.

In the financial field, many researchers have devoted themselves to the study of point, interval and probabilistic forecast. For example, Setiawan et al. [24] built long short-term memory (LSTM) network to process time series data and make point forecast of stock prices. Kovacs et al. [25] used recurrent neural networks (RNN) to model the stock market and achieve the corresponding point forecast. Tashakkori et al. [26] explored the application of RNN and LSTM networks for stock price prediction over a 10-day horizon. Xiong et al. [27] used multi-output support vector regression to make interval forecast for long and short term stock prices. Jiang et al. [28] proposed a dual convolutional neural network to predict the range of stock prices, and conducted extensive experiments. Uniejewski and Weron [29] proposed a quantile regression regularization variant for probabilistic forecast of electricity prices, which performs well in both Polish and Nordic electricity market data sets. Cao et al. [30] built a new probabilistic forecast model, called quantile time convolutional network, which can accurately describe the uncertain fluctuations of carbon prices.

In contrast, the purpose of scenario generation is to present uncertain information by a set of several deterministic scenarios. Since scenario generation can effectively reduce risks and assist decision-making, this method has been successful used in some fields other than finance. Yu et al. [31] proposed a deep convolutional network based on spatiotemporal characteristics to predict wind power, where wind turbines are used to collect data at every moment, and time series of scenarios within a period of time are taken as the characteristics. In addition, GAN-based methods have been used to make scenario generation. Han et al. [32] proposed a GAN-based remote sensing image generation method, which can be used to annotate samples with high resolution for scenario classification. Specifically, the wasserstein distance measures the difference between the actual distribution and the generator distribution, thereby addressing the issue of gradient disappearance during sample generation. Engemann and Lessmann [33] adopted an approach based on a conditional WGAN which can effectively model tabular datasets with numerical as well as categorical variables, with special attention to downstream classification tasks through auxiliary classifier loss. The traditional prediction method is to firstly generate the scenarios and then make forecast through modeling. Chen et al. [34] proposed a GAN-based data-driven scenario generation method, whereafter a constrained optimization problem is

Table 1
Advantages and disadvantages of different methods.

Tasks	Methods	Advantages	Disadvantages
Time series forecasting	RNN	Shared weights	Vanishing/exploding gradients
	LSTM	Mitigation of vanishing/exploding gradients	Hyperparameter sensitivity
	Transformer	Strong long-term dependency modeling	High computational and spatial complexity
Correlation extraction	CNN	Invariance to transformations	Lack of interpretability
	DNN	Universal approximators	Vanishing/exploding gradients
	GCN	Efficient feature extraction	High computational cost
Scenario generation	HMM	Simplicity and interpretability	Limited to short-range dependencies
	VAE	Continuous latent space	Blurry reconstructions
	WGAN	Improved training stability	Complex implementation

formulated to achieve scenario forecasts. While current methods can make end-to-end forecasts directly. Li et al. [35] used WGAN to study large amounts of collision data to forecast the severity of injuries caused by traffic accidents.

Besides, in terms of the input information used, traditional stock price forecasting methods are mainly based on historical stock data. For example, Singh and Borah [36] proposed a model based on type-2 fuzzy time series and particle swarm optimization, then used SBI as well as Google stock index data set to test and verify, showing the superiority of this model in comparison with other baselines. Sangeetha and Alfia [37] used machine learning technique based on evaluative linear regression to predict stock financial values of the standard and poor's 500 (S & P 500) index, including open, close, low, high, as well as volume factors.

With the development of machine learning technology, researchers have gradually used different modal information to make forecasting. For example, Li et al. [38] used the sentiment dictionary to evaluate the positivity or negativity of news texts, then connected it with the ups or downs of historical stock price data, and realized the trend forecast through SVM. Gao et al. [39] calculated the sentiment score of related news articles through the Loughran-McDonald sentiment dictionary, then obtained the sentiment index of news providers, and finally used RNN to predict the trend of stock prices. Zhang et al. [40] established the attention mechanism-based Transformer model to combine stock price historical data with social media texts to forecast stock price trends.

In addition to considering cross-models, correlations among different stocks or companies are also key factors in stock price forecasting. Nam and Seong [41] considered the causal relationship among companies, using transfer entropy to find such a causal relationship, whereafter multi-core learning was applied to combine the characteristics of target companies and related companies. Through testing the Korean market data set, it is found that this method is superior to the traditional ones. Ma et al. [42] combined the correlation among different stocks and used bi-directional LSTM to predict the rise or fall.

In the past, there have been many related studies on modeling in the financial field. Fadavi [43] examined the sub-game perfect Nash equilibrium of price competition within an oligopoly market where goods are perishable assets. Alahdadi et al. [44] proposed a truthful incentive double auction mechanism with paying rewards in the cloud environment.

In order to more intuitively compare the applicability of different approaches, we have listed the advantages and disadvantages of different methods in Table 1. In view of the temporal-spatial correlation of stocks, and the complicated uncertainty of stock market, we use WGAN, GCN and Transformer to construct the stock price forecasting model in this paper.

Furthermore, a comprehensive comparison of the present study to the existing stock price forecasting literature is presented in Table 2 to distinguish the difference and highlight the superiority of our method.

As the above literature review shows, stock price forecasting is a very concerned problem, its technical methods are constantly improving and developing.

3. Basic knowledge

This chapter delves into some fundamental knowledge, encompassing the notion of stock scenario generation alongside the algorithmic foundations of WGAN, Transformer as well as GCN.

3.1. Scenario generation of stock price

Scenario generation refers to the use of techniques such as statistics analysis and machine learning to visualize the possible future time series.

In the field of stock price forecasting, scenario generation is to provide multiple time series, and each time series represents a situation of the future stock price. By generating multiple forecast scenarios, the uncertainty and risk of stock prices can be fully assessed. Fig. 1 shows the visualization of point forecast, interval forecast, probabilistic forecast and scenario generation in stock price forecasting.

We observe that the outcomes of the point forecast frequently diverge from the actual value. Although interval forecast uses upper and lower levels to bound the future stock price, it is short of probabilistic information, resulting in a lack of statistical basis for decision making. Besides, it is difficult to have an accurate probability distribution to fit the uncertainty, and probability distribution is also difficult to apply to time series because it is usually carried out independently in time segments. In contrast, scenario generation uses historical data and other related information to generate a number of deterministic stock price sequences, then employs a set of multiple sequences to represent uncertainty. Compared with interval forecast and probabilistic forecast, the generated scenarios can be directly used in portfolio [45] optimization.

3.2. Wasserstein GAN

GAN is a deep learning model that works by using two neural networks against each other to generate realistic data such as images, audio or text, and its loss function is generally cross-entropy loss.

WGAN is a variant of GAN, and its main innovation is the introduction of the Wasserstein distance [46] to measure the difference between the distribution generated by the generator and the distribution of the real sample, solving the training as well as stability problems in traditional GANs.

Since the goal of WGAN is to minimize the Wasserstein distance between the generator and the discriminator, instead of using cross-entropy loss or Kullback-Leibler divergence, the following Wasserstein distance is used:

$$W(P_r, P_g) = \inf_{\gamma \sim \Pi(P_r, P_g)} E_{(x,y) \sim \gamma} [\|x - y\|] \quad (1)$$

where P_r represents the real data distribution, P_g represents the output distribution of the generator, and $\Pi(P_r, P_g)$ represents all possible joint distributions such that the edge distributions are the set of P_r and P_g respectively. In order to achieve the optimization of WGAN, the loss functions of generator and discriminator are as follows:

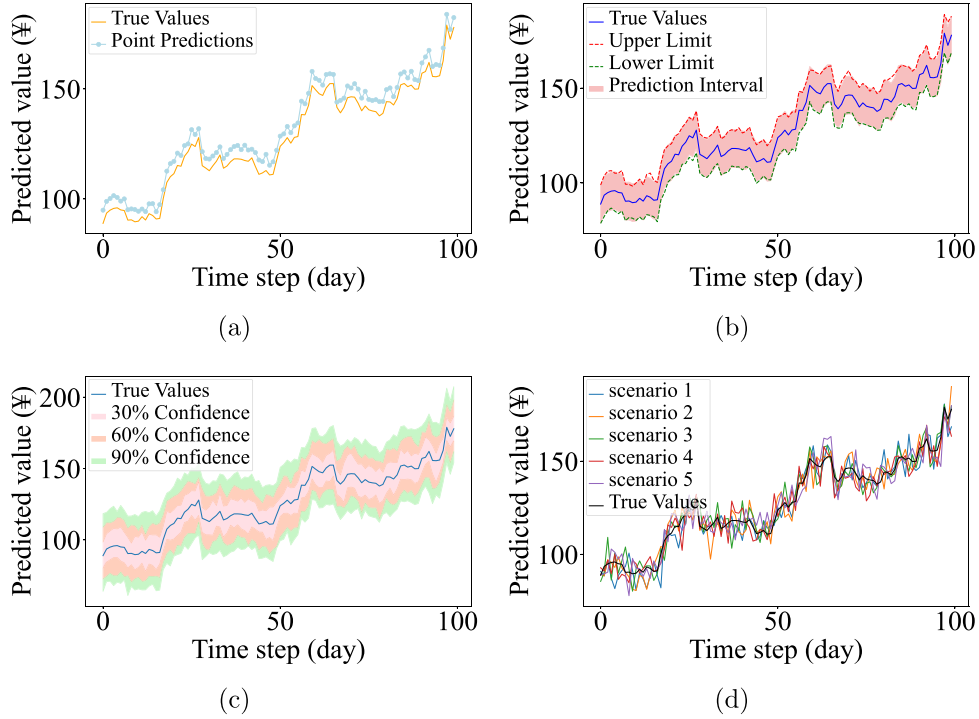
$$G_{loss} = \min_G E_{z \sim p_z} [W(P_r, P_g)] \quad (2)$$

Table 2

Comparison of this paper to ten existing stock price forecasting studies.

Criteria	Setiawan et al. [24]	Kovacs et al. [25]	Tashakkori et al. [26]	Xiong et al. [27]	Jiang et al. [28]	Li et al. [38]	Gao et al. [39]	Zhang et al. [40]	Nam and Seong [41]	Ma et al. [42]	Ours
Complex model	✓	✓	✓	×	✓	×	✓	✓	×	✓	✓
Cross-modal information	×	×	×	×	×	✓	✓	✓	✓	✓	✓
Stock correlation	×	×	×	×	✓	×	×	×	✓	✓	✓
Scenario generation	×	×	×	×	×	×	×	×	×	×	✓
Multiple statistical indicators	✓	×	×	✓	✓	✓	✓	✓	✓	✓	✓
Real market performance	×	×	×	✓	×	×	✓	✓	×	✓	✓

✓: criteria fully satisfied, ×: criteria not satisfied.

**Fig. 1.** Visualization of four forecasting methods: (a) Point forecast, (b) Interval forecast, (c) Probabilistic forecast, and (d) Scenario generation.

$$D_{loss} = \max_D \left(E_{x \sim p_{data}} [D(x)] - E_{z \sim p_z} [D(G(z))] \right) \quad (3)$$

Fig. 2 shows the structure of WGAN. By introducing Wasserstein distance to measure the difference between distributions, WGAN improves the training stability of the generator, makes it easier to generate high-quality samples, and solves the problem of unstable training existing in traditional GAN.

3.3. Transformer

The attention mechanism is the core of Transformer [47], which can be regarded as an addressing operation: the input information $\{x_1, x_2, \dots, x_n\}$ is regarded as the contents stored in the memory, the elements are composed of *Key* and *Value*, the current *Key* = *Query* query is aimed at retrieving the corresponding *Value* in the memory, that is, the attention value.

(1) Calculation of the correlation degree

For each input element x_i , we calculate its correlation score $score_i$ with the query item q :

$$score_i = f(q, x_i) \quad (4)$$

Here f is a learnable function, which can be a dot product, a linear transformation, a neural network, etc.

(2) Calculation of the attention weight

The attention weight distribution is obtained by normalizing the correlation degree score:

$$attention_i = softmax(score_i) = \frac{\exp(s_i)}{\sum_{j=1}^n \exp(s_j)} \quad (5)$$

The *softmax* function makes the sum of the attention weights equal to 1, so that the weights of different elements reflect their relative importance.

(3) Calculation of the weighted representation

The input elements are weighted and summed using the attention weights to get a weighted representation:

$$attention((K, V), Q) = \sum_{i=1}^N attention_i V_i \quad (6)$$

$$= \sum_{i=1}^N \frac{\exp(s(k_i, q))}{\sum_{j=1}^n \exp(s(k_j, q))} V_i$$

$$attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

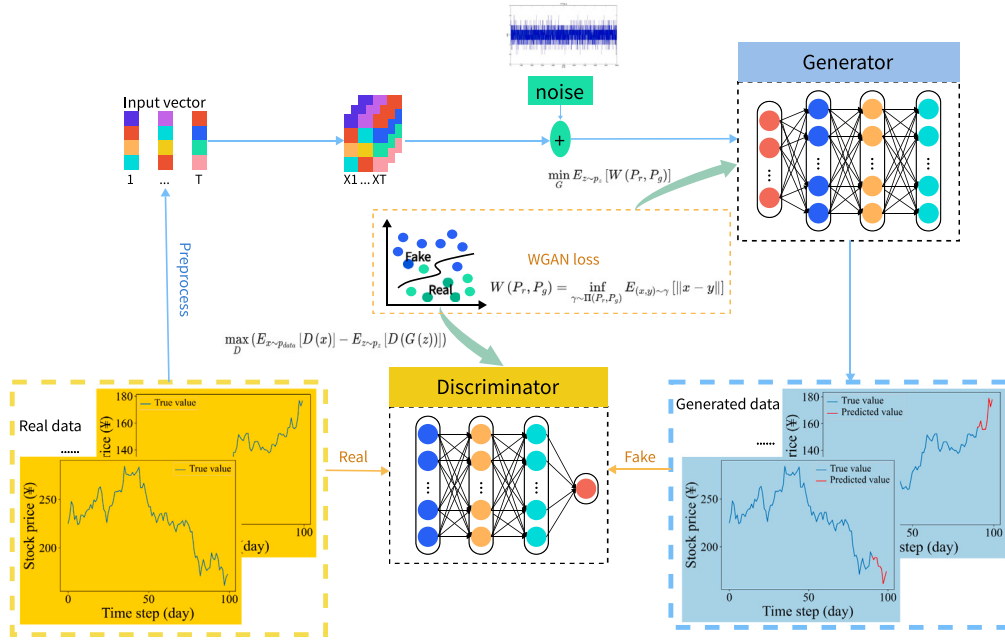


Fig. 2. Detailed structure of WGAN.

To better capture different relationships, Transformer introduces multiple self-attention heads. Each head has a different Q , K , and V weight matrix, which is finally spliced and linearly transformed to obtain the output of multi-head attention.

The Transformer model does not have explicit sequence position information, so positional encoding needs to be added to bring word position information into the model:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad (8)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad (9)$$

To avoid disappearing gradients, residual joins as well as layer normalization are applied after each sub-layer, such as the self-attention layer and the feedforward neural network layer. The output can be expressed as:

$$Y = LayerNorm(X + Sublayer(X)) \quad (10)$$

Where $Sublayer(X)$ is the output of the sublayer.

In each encoder and decoder layer, a feedforward neural network is included for nonlinear transformation. Feedforward neural networks usually consist two linear transformations and an activation function, such as ReLU.

$$FFN(X) = \max(0, XW_1 + b_1)W_2 + b_2 \quad (11)$$

Generally, the encoder and decoder consist multiple layers, while the output of the encoder acts as the input of the decoder, the decoder considers the previously generated tags autoregressively when generating each output tag. Both the encoder and the decoder need masks. The function of the encoder's mask is to remove the influence of various padding in the training process. Besides this function, the decoder's mask can also cover the input to prevent the decoder from seeing the things to be predicted later.

The output layer maps the final hidden representation of the model to a probability distribution on the vocabulary. Usually linear transformations and functions are used to calculate the probability distribution of words in each vocabulary.

3.4. Graph convolution network

As a neural network model for processing graph-structured data, GCN performs well in tasks such as node classification and graph generation. The core idea of GCN is to update the representation of nodes by using the neighbor information, so as to capture the structural information among nodes.

Suppose there is a graph $G = (V, E)$ where V represents the set of nodes and E represents the set of edges. Each node v_i has an eigenvector x_i , while each edge (v_i, v_j) represents the connection between nodes v_i and v_j . The core formula of GCN can be expressed as:

$$H^{(l+1)} = \sigma\left(D^{-\frac{1}{2}} \tilde{A} D^{-\frac{1}{2}} H^{(l)} W^{(l)}\right) \quad (12)$$

where $H^{(l)}$ represents the embedding of the l layer, $W^{(l)}$ is the linear transformation matrix of the l layer, A represents the adjacency matrix, D represents the degree matrix of A , $\tilde{A}$ represents the adjacency matrix with self-ring, and σ is an activation function.

The above formula shows that the representation of a node is obtained by a linear combination of the representations of its neighbors, where $\tilde{A}H^{(l)}$ represents a weighted summation of the neighbor features of node v_i , with the weights determined by the adjacency matrix A . $D^{-\frac{1}{2}}$ is the reciprocal of the square root of the degree matrix used to normalize the weights of neighbors.

GCN can gradually expand the representation capability of nodes by stacking multiple layers, and the network can gradually capture the higher-order structure information of nodes by updating the node representation several times.

4. The proposed method

As shown in Table 1 in Section 2, Transformer stands out for its exceptional ability to handle long-range dependencies in sequences. Meanwhile, GCN excels in performing efficient feature extraction from graph-structured data. WGAN stands out for its notable stability during training. By combining these three powerful models, we are able to leverage their unique strengths to address a wide range of challenges in our method.

Therefore, in this section, a novel cross-modal scenario generation (CM-SG) method for stock price forecasting is proposed. Especially, the

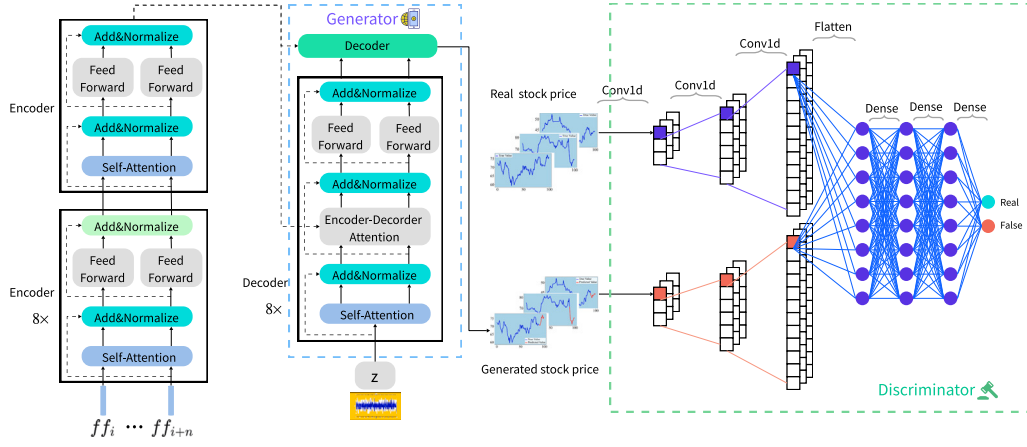


Fig. 3. Internal structure of the scenario generation portion.

Table 3
Nomenclature.

Notation	Description
$o_{i,j}$	Opening price of the j stock on day i
$c_{i,j}$	Closing price of the j stock on day i
$h_{i,j}$	Highest price of the j stock on day i
$l_{i,j}$	Lowest price of the j stock on day i
num_i	Number of news headlines on day i
T	Total number of days of data
seq	Length of the word encoding
x_p	The p stock associated with the target stock
$x_{0,i}$	On day i , the characteristic representation of the relevant stock to the target stock
$n_{i,num}$	The num headline on day i
$z_{i,num}$	Word count for the num headline on day i
$k_{i,num,p,q}$	The q code of the p word of the num headline on day i
ω_{num}	Number of reads for headline num
$day_{m,n}$	Average of the n headline codes on day m , $day_{m,n} \in R^{1 \times k}$, k represents the encoding length
n'_m	Weighted eigenvector on day m , $n'_m \in R^{1 \times k}$
n''_m	The eigenvector after dimensionality reduction on day m , $n''_m \in R^{1 \times k}$
df_i	Data characteristics of day i , $df_i \in R^{1 \times data}$, $data$ represents the data feature dimension
nf_i	News features of day i , $nf_i \in R^{1 \times text}$, $text$ represents the text feature dimension
$S_{m,n}$	Similarity matrix, $S_{m,n} \in R^{data \times text}$
$\alpha_{m,n}$	Attention weight
ff_i	Fusion vector on day i , $ff_i \in R^{1 \times fusion}$

structured historical data and unstructured market news are considered to improve the forecast performance, while GCN is applied to obtain the representation of target stock features. Finally, the scenario generation of future stock price is achieved by using WGAN. The detailed structure of scenario generation is shown in Fig. 3. And the overall flowchart of CM-SG is shown in Fig. 4, which is divided into four parts: target stock feature extraction, related stock feature extraction, feature fusion and scenario generation. Decoder of Transformer model is used as WGAN's generator and convolutional neural network (CNN) is used as discriminator. Additionally, this section details the methods we used to verify the effectiveness of the model. The meanings of symbols appearing in the formula are shown in Table 3.

4.1. Target stock features preprocessing

4.1.1. Data feature preprocessing

Similar to most literature, for example Liang et al. [48], we select four kinds of stock features, namely the opening price (o), closing price (c), highest price (h) and lowest price (l), which form the 1×4 feature vectors $\{o, c, h, l\}$ for each trading day. In total, T days are

selected as the time window to predict the closing price after q days. All eigenvectors of T days are normalized to improve the convergence speed, performance and stability of the subsequent model.

4.1.2. Text feature preprocessing

It has been proved that a news headline can effectively summarize the content and emotion of the whole article [49], so we crawled all the news headlines of the target news of T days from website. These headlines are not equally important, and theoretically speaking, the more widely the news headlines have been read, the greater their impact on the future stock price will be. Therefore, the following preprocessing is performed on the news headlines.

(1) Put all the headlines in chronological order, with the ones of the same date together:

$$\{n_{i,1}, n_{i,2}, \dots, n_{i,num_i}\}, \{n_{i+1,1}, n_{i+1,2}, \dots, n_{i+1,num_{i+1}}\}, \dots, \{n_{i+T-1,1}, n_{i+T-1,2}, \dots, n_{i+T-1,num_{i+T-1}}\} \quad (13)$$

where i is the date and num_i is the number of news headlines on day i .

(2) Use pre-trained model to encode all words of all headings for T days, assuming the encoder is represented by F , for example:

$$F(n_{i,1}) = \{k_{i,1,1,1}, k_{i,1,1,2}, \dots, k_{i,1,1,seq}\}, \dots, \{k_{i,1,z_{i,1},1}, k_{i,1,z_{i,1},2}, \dots, k_{i,1,z_{i,1},seq}\} \quad (14)$$

$$F(n_{i,2}) = \{k_{i,2,1,1}, k_{i,2,1,2}, \dots, k_{i,2,1,seq}\}, \dots, \{k_{i,2,z_{i,1},1}, k_{i,2,z_{i,1},2}, \dots, k_{i,2,z_{i,1},seq}\} \quad (15)$$

Among them, the four corner symbols represent the date, the number of the title, the number of words, and the code number, that is, $k_{i,1,1,1}$ represents a number.

(3) Make an average of each headline for each day, that is, treat each news headline as an average word:

$$day_{m,n} = \frac{1}{z_{m,n}} \sum_{h=1}^{z_{m,n}} (\{k_{m,n,h,1}, k_{m,n,h,2}, \dots, k_{m,n,h,seq}\}) \quad (16)$$

where m and n represent the date and the number of news headlines, respectively. Then make a weighted average according to the amount of reading ω , that is, take all the news of each day as a word, and use a $1 \times seq$ vector to represent it:

$$n'_m = \frac{\sum_{i=1}^{num} \omega_i day_{m,i}}{\sum_{i=1}^{num} \omega_i} \quad (17)$$

The dimensionality of n'_m is reduced by a linear layer:

$$n''_m = Linear(n'_m) \quad (18)$$

Finally, the target stock news features of T days are obtained.

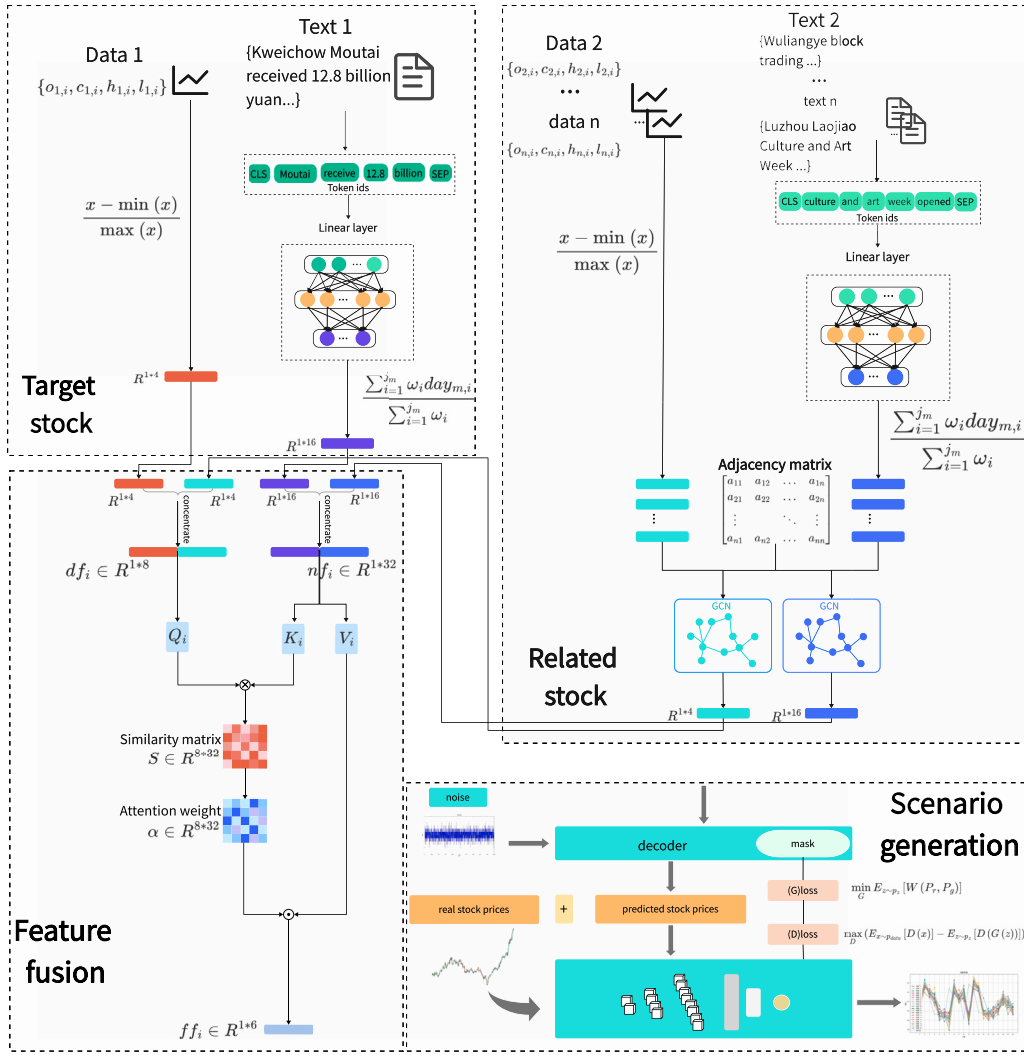


Fig. 4. Flowchart of CM-SG.

4.2. Stock correlation feature extraction

Since correlations among different stocks do exist, it is not enough to consider only the target stock, but also several stocks related to the target stock. According to Pearson coefficient, p branch stock $\{x_1, x_2, \dots, x_p\}$ which is most related to the target stock x_0 is selected, and adjacency matrix A is constructed.

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,p+1} \\ a_{2,1} & a_{2,2} & \dots & a_{2,p+1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p+1,1} & a_{p+1,2} & \dots & a_{p+1,p+1} \end{bmatrix} \quad (19)$$

Where $a_{i,j} = 1$ if x_i is related to x_j , and otherwise $a_{i,j} = 0$.

4.2.1. Data feature correlation extraction

We select a few stocks that are most relevant to the target stocks, so they are fully connected, and each element in the matrix is set to 1. The adjacency matrix is input into GCN, and the data features of the target stocks are updated every day through convolution operations. We choose GCN because, compared with traditional graph neural network (GNN), GCN can make better use of local neighborhood information, so that the node representation is more locally consistent and more efficient. Consider on day i :

$$\{x_{1,i}, x_{2,i}, \dots, x_{p,i}\} = \{o_{1,i}, c_{1,i}, h_{1,i}, l_{1,i}\}, \{o_{2,i}, c_{2,i}, h_{2,i}, l_{2,i}\}, \dots, \{o_{p,i}, c_{p,i}, h_{p,i}, l_{p,i}\} \quad (20)$$

$$\begin{cases} x_{0,i} = GCN(x_{1,i}, x_{2,i}, \dots, x_{p,i}) \\ x_{0,i+1} = GCN(x_{1,i+1}, x_{2,i+1}, \dots, x_{p,i+1}) \\ \dots \\ x_{0,i+T-1} = GCN(x_{1,i+T-1}, x_{2,i+T-1}, \dots, x_{p,i+T-1}) \end{cases} \quad (21)$$

4.2.2. Text feature correlation extraction

The news characteristics of p stocks are obtained in accordance with Section 4.1.2.:

$$\begin{cases} \{n''_{1,i}, n''_{1,i+1}, \dots, n''_{1,i+T-1}\}, \{n''_{2,i}, n''_{2,i+1}, \dots, n''_{2,i+T-1}\} \\ \dots, \{n''_{p,i}, n''_{p,i+1}, \dots, n''_{p,i+T-1}\} \end{cases} \quad (22)$$

Similarly, through the GCN, the news features of the p stocks are used to update the news features of the target stocks every day. Since these stocks are all related to each other, the adjacency matrix is also set to 1.

Through convolution operation, the news feature of target stock can be represented by the news feature of p stocks:

$$\begin{cases} n''_{0,i} = GCN(n''_{1,i}, n''_{2,i}, \dots, n''_{p,i}) \\ n''_{0,i+1} = GCN(n''_{1,i+1}, n''_{2,i+1}, \dots, n''_{p,i+1}) \\ \dots \\ n''_{0,i+T-1} = GCN(n''_{1,i+T-1}, n''_{2,i+T-1}, \dots, n''_{p,i+T-1}) \end{cases} \quad (23)$$

Finally, the target stock news features combined with relevant stock features of T days are obtained.

4.3. Feature fusion based on attention mechanism

After the both two features are obtained, both of them are connected in series according to the corresponding days to obtain new data features:

$$df_i = \text{concat}(\{o, c, h, l\}_i, x_{0,i}), df_i \in R^{1 \times 8} \quad (24)$$

Then the two text features are connected in series according to the corresponding days to get the new news features:

$$nf_i = \text{concat}(n''_i, n''_{0,i}), nf_i \in R^{1 \times 32} \quad (25)$$

Then, based on the attention mechanism, the feature fusion of df_i and nf_i is carried out. For the fusion of data feature and text feature, in fact, based on the idea of attention mechanism, first multiply the data feature (1×8) by the transpose of the text feature (1×32) to obtain a matrix (8×32), then do the *softmax* operation on the matrix, finally multiply this matrix by the text features, and take the first six numbers as the fusion features. Let us use some formulas to make it more specific:

Firstly, the similarity matrix S is calculated:

$$S_{m,n} = \frac{df_{i,m} * nf_{i,n}}{\|df_{i,m}\| * \|nf_{i,n}\|}, m = 1, 2, \dots, 8, n = 1, 2, \dots, 32 \quad (26)$$

Next calculate the attention weight between df_i and nf_i :

$$\alpha_{m,n} = \frac{\exp(S_{m,n})}{\sum_{k=1}^{32} \exp(S_{m,k})} \quad (27)$$

Thus, the fused eigenvector $f f_i$ is obtained.

$$f f_{i,k} = \sum_{j=1}^{32} \alpha_{k,j} n f_{i,j}, k = 1, 2, \dots, 6 \quad (28)$$

Finally, the fusion characteristics of T days are obtained.

4.4. Scenario generation

Different from point forecast or trend forecast in previous studies, this paper uses GAN to achieve scenario generation, which is effective for investors to comprehensively perceive risks, so as to make proper investment decisions.

GAN includes generator G and discriminator D . In this paper, Wasserstein GAN is used for scenario generation, Transformer is used as the WGAN generator, that is, forecasting model, and CNN is used as the WGAN discriminator to distinguish generated scenarios from real scenarios.

From the Gaussian noise distribution γ , k samples are randomly sampled, namely $\{z_1, z_2, \dots, z_k\}$, and then randomly sample k samples from the real data distribution $\{close_i, close_{i+1}, \dots, close_{i+T-1}\}$, that is $\{x_1, x_2, \dots, x_k\}$.

The parameters of discriminator D were updated by small-batch random gradient descent. The specific formula is as follows:

$$\nabla \frac{1}{k} \sum_{i=1}^k [D(x_i) - D(G(z_i))] \quad (29)$$

Then, another k samples are randomly sampled from Gaussian noise distribution γ , and the parameters of generator G are updated by small-batch random gradient descent method. The specific formula is as follows:

$$-\nabla \frac{1}{k} \sum_{i=1}^k D(G(z_i)) \quad (30)$$

After obtaining the future scenarios of stock prices, certain loss functions need to be introduced to judge the performance of the forecast results. There are three kinds of loss functions used in this paper,

namely mean square error (MSE), root mean square error (RMSE) and mean absolute percentage error (MAPE):

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2 \quad (31)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2} \quad (32)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{x}_i - x_i}{x_i} * 100\% \right) \quad (33)$$

In addition, we set up a CIR indicator that represents the proportion of the true value in the 95% confidence interval.

4.5. Portfolio selection model

Finally, it is necessary to further verify the effectiveness of scenario generation by using real-life investment problems. Therefore, the following portfolio selection problem based on the classical mean-variance model [50] is considered:

$$\begin{cases} \max_{\omega_i} E(r_p) = \sum_{i=1}^k \omega_i r_i \\ \text{s.t.} \\ \text{Var}(r_p) = \sum_{i=1}^k \omega_i^2 \text{Var}(r_i) + \sum_{i \neq j} \omega_i \omega_j \text{Cov}(r_i, r_j) \leq r \\ \sum_{i=1}^k \omega_i = 1, \omega_i \geq 0 \end{cases} \quad (34)$$

where r_p is the return rate of the portfolio, r_i is the return rate of a single stock, ω_i is the investment proportion on each stock, k is the number of stocks we selected.

In order to comprehensively consider the risks brought by investment, some additional constraints are added: (1) The probability of non-negative return rate is greater than or equal to α . (2) The lowest possible yield is greater than or equal to β . In addition, the resulting scenario generation results are entered into it. The goal is to choose the weight that maximizes our expected return. And it can be viewed as a combinatorial optimization problem:

$$\begin{cases} \max_{\omega_i} \sum_{i=1}^k \omega_i \sum_{j=1}^{t_i} \Pi_{i,j} S_{i,j} \\ \text{s.t.} \\ \text{Var}(S) = \sum_{i=1}^k \omega_i^2 \text{Var}(S_i) + \sum_{i \neq j} \omega_i \omega_j \text{Cov}(S_i, S_j) \leq r \\ P(S \geq 0) \geq \alpha \\ \sum_{i=1}^k \omega_i * \min(S_i) \geq \beta \\ \sum_{i=1}^k \omega_i = 1, \omega_i \geq 0 \end{cases} \quad (35)$$

where t_i is the number of scenarios obtained by the i stock, $S_{i,j}$ is the return rate of scenarios, i is the stock serial number, j is the scenario serial number, and $\Pi_{i,j} = \frac{1}{t_i}$ is the probability of the occurrence of scenarios.

Moreover, the constraint levels r , α , β can be updated without difficulty to meet the requirements of different types of investors, thus obtaining various trade-offs between investment return and risk.

5. Experimental result

In this section, the effectiveness of the proposed CM-SG is justified by a number of experiments.

The stock history data in this paper is collected from <https://www.yahoo.com> and the news headlines are from <https://www.eastmoney.com>. Code implementation and case experiments were carried out on a Jupyter Notebook (anaconda) with Intel i5-10400F 16 GB CPU and NVIDIA GeForce GTX 1660 6 GB GPU.

The tuning method of Bayesian optimization [51] is used to adjust the network hyperparameters in this paper, and finally some hyperparameters are set as follows:

The generator has 8 Decoder layers, each layer's dropout is 0.2, the activation function uses ReLU function, the learning rate is 0.001, the batchsize is 32. The ratio of training set, verification set and test set is 8:1:1 (2000 days, 250 days, 250 days, respectively), training 100 epochs. The discriminator consists of 3 Conv1D layers, 1 Flatten

Table 4
Symbol, name, and main business of the selected stock.

No.	Stock ID	Name	Main business
1	600519	GZMT	Production and sale of wine
2	000538	YNBY	R&d, production and sales of pharmaceutical products
3	601988	ZGYH	Corporate finance, personal finance and financial markets
4	000063	ZXTX	Provide customers with satisfactory ICT products and solutions
5	002739	WDDY	Production and distribution of television series and films
6	300002	SZTY	Internet operation services and mobile information security services
7	603288	HTWY	Condiment production enterprise
8	002233	TPJT	All kinds of Portland cement production and sales
9	000651	GLDQ	Air conditioning, refrigerator and other products R & D, production and sales
10	002594	BYD	Automobile, rail transit, new energy and electronics
11	600009	SHJC	Aviation ground support services and supporting services
12	601888	ZGZM	Wholesale, retail and other business of duty-free goods

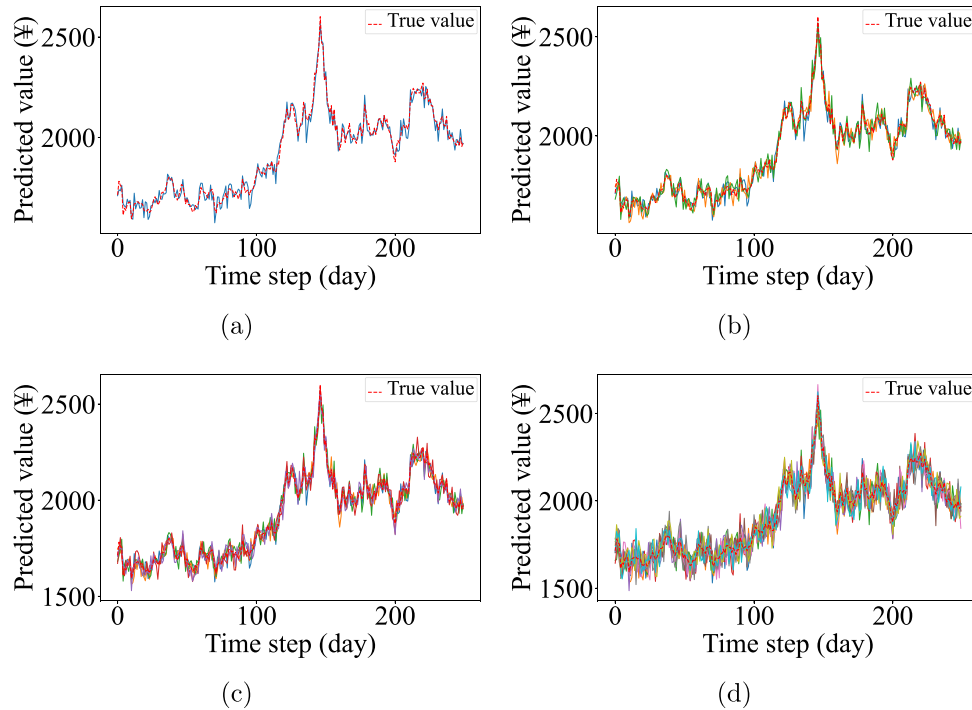


Fig. 5. Scenario generation results: (a) 1 scenario (point forecast), (b) 3 scenarios, (c) 5 scenarios, and (d) 10 scenarios.

layer and 3 Dense layers, while the activation function uses the ReLU function with a learning rate of 0.01.

We select several different types of stocks whose knowledge is shown in Table 4, and then carry out scenario generation for these stocks respectively by using CM-SG.

Then, we set up the subsequent experiments to showcase the efficacy of the suggested model.

5.1. Scenario generation of CM-SG

We calculate the pearson correlation coefficient (PCC) among the closing prices of all chinese stocks and select 10 stocks with the highest correlation coefficient with the target stocks. Take Kweichow Moutai (600519) as an example, 10 stocks are selected, whose information and the correlation coefficient with Kweichow Moutai are shown in Table 5 (ranked in a descend order). The stock price data of these stocks are obtained from July 1, 2015 to July 1, 2023, including the open, close, high and low prices. Among them, the missing values are dealt with in such a way that if a day is a non-trading day, the data characteristics of that day are consistent with the previous trading day.

Then we crawl the news headlines and the corresponding reading volume of the news of these stocks from July 1, 2015 to July 1, 2023. If there is no information about a particular stock on a given day, the

Table 5
Correlation coefficients of relevant stocks.

Code	Name	Correlation coefficient
000568	LZLJ	0.891
002304	YHGF	0.855
600809	SXFJ	0.792
600436	PZH	0.781
000858	WLY	0.771
300408	SHJT	0.715
000596	GJGJ	0.711
600887	YLGF	0.673
600559	LBGJ	0.669
002080	ZCKJ	0.626

news headlines and reading volume for that day remain the same as the previous day. For the coding of the text, since chinese titles were selected, the encoder selected in this paper is roberta-base-finetuned-jd-full-chinese encoder from <https://github.com/huggingface>.

To more directly illustrate the experiment results, the scenario generation of CM-SG with 1, 3, 5 and 10 scenarios are given in

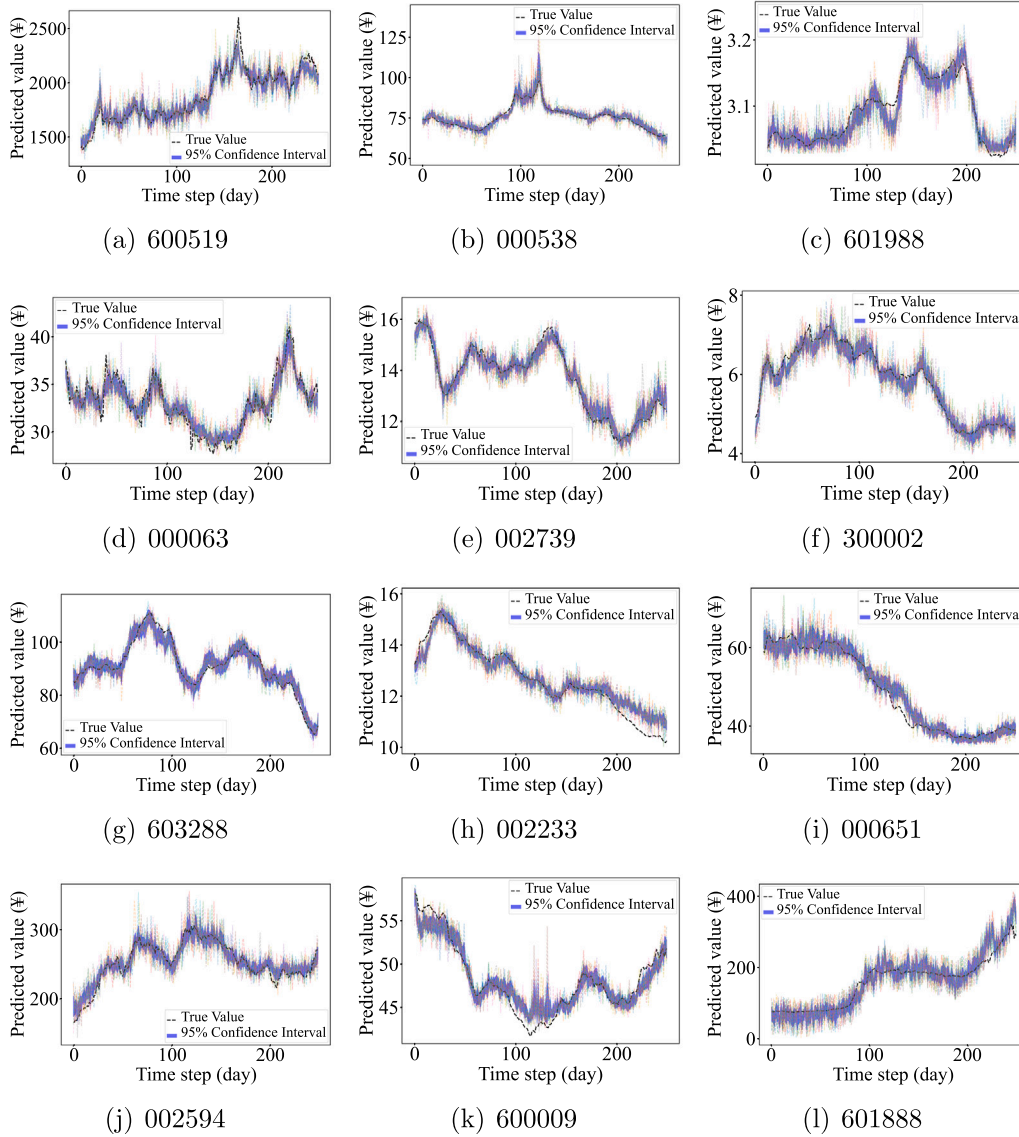


Fig. 6. Scenario generation results with 95% confidence interval for all 12 stocks.

Fig. 5(a)–(d). As can be seen from the figure, the real value of stock price is mostly covered by the result of scenario generation.

Next, we used the proposed CM-SG to generate 10 scenarios for all 12 stocks and plot the 95% confidence interval. The results are shown in Fig. 6. Intuitively, the rise and fall trend of the scenarios is basically synchronized with the real value.

5.2. Comparison with baseline methods

In order to verify the effectiveness of the method presented in this paper, three baseline methods are introduced: CH-RNN [52], GRU [53] and E-LSTM [54]. The average MSE and MAPE are used as evaluation indicators. The scenario generation results and the PCC visualization of different methods are shown in Figs. 7 and 8, while Table 6 gives the scenario evaluation on MSE and MAPE.

It can be seen that the performance of CH-RNN is relatively poor, while E-LSTM and GRU can achieve the minimum error in 4 stocks, and their effect is slightly better than CH-RNN. Besides, in most cases, that is 8 stocks, our method can achieve the minimum error and obtain the best forecast results. Compared with GRU, our method achieves an average reduction of 12.41% in MSE and 7.27% in MAPE. And

compared with E-LSTM, the improvements are 13.54% and 9.59% respectively. These demonstrate the excellence of our method.

In addition, we calculated and analyzed CIR indicators of different methods in Table 7.

As can be seen from the results, our approach is the best in the performance of the CIR indicator among 8 stocks, followed by GRU and E-LSTM, which perform the best in 3 stocks and 2 stocks respectively, while CH-RNN performed the worst.

Furthermore, we give the number of parameters and training time of different methods in Table 8.

5.3. Ablation experiments

The proposed model CM-SG includes several parts: scenario generation, cross-modal feature fusion and stock correlation. In order to verify the effectiveness of each part on the forecasting accuracy, we set three sets of ablation experiments, which are described in Table 9. The average MSE and MAPE of the five stocks were calculated for each group of ablation experiments, while the results are also shown.

It can be seen that our method consistently outperforms others, demonstrating the effectiveness of correlation, feature fusion, and scenario generation in enhancing forecasting accuracy.

Table 6
Scenario evaluation of different methods on MSE and MAPE.

Methods	600519		000538		601988		000063	
	MSE	MAPE	MSE	MAPE	MSE	MAPE	MSE	MAPE
CH-RNN	10 773.3	4.47%	53.22	6.74%	0.0129	2.96%	8.69	6.02%
GRU	7907.36	3.66%	41.49	6.03%	0.0097	2.31%	5.58	5.49%
E-LSTM	6127.44	3.13%	41.12	5.97%	0.0110	2.63%	5.66	5.54%
Ours	5364.12	3.05%	43.61	7.33%	0.0090	2.21%	4.76	5.16%
Methods	002739		300002		603288		002233	
	MSE	MAPE	MSE	MAPE	MSE	MAPE	MSE	MAPE
CH-RNN	4.2139	9.39%	0.2612	7.64%	71.7781	8.21%	0.86	5.28%
GRU	1.6865	6.22%	0.2448	6.90%	67.0664	7.89%	0.50	4.34%
E-LSTM	2.0335	6.57%	0.2528	8.06%	55.2616	6.76%	0.65	4.73%
Ours	1.9393	6.45%	0.2120	6.57%	50.3305	6.43%	0.45	4.09%
Methods	000651		002594		600009		601888	
	MSE	MAPE	MSE	MAPE	MSE	MAPE	MSE	MAPE
CH-RNN	48.8341	11.71%	1134.47	13.38%	74.1686	11.77%	2475.69	19.07%
GRU	23.8598	7.18%	847.32	10.32%	40.0358	8.16%	1855.95	15.76%
E-LSTM	21.9969	7.01%	1292.73	14.17%	33.8232	7.58%	1502.66	14.31%
Ours	17.1788	7.96%	1034.54	11.42%	30.6905	7.52%	1252.41	13.24%

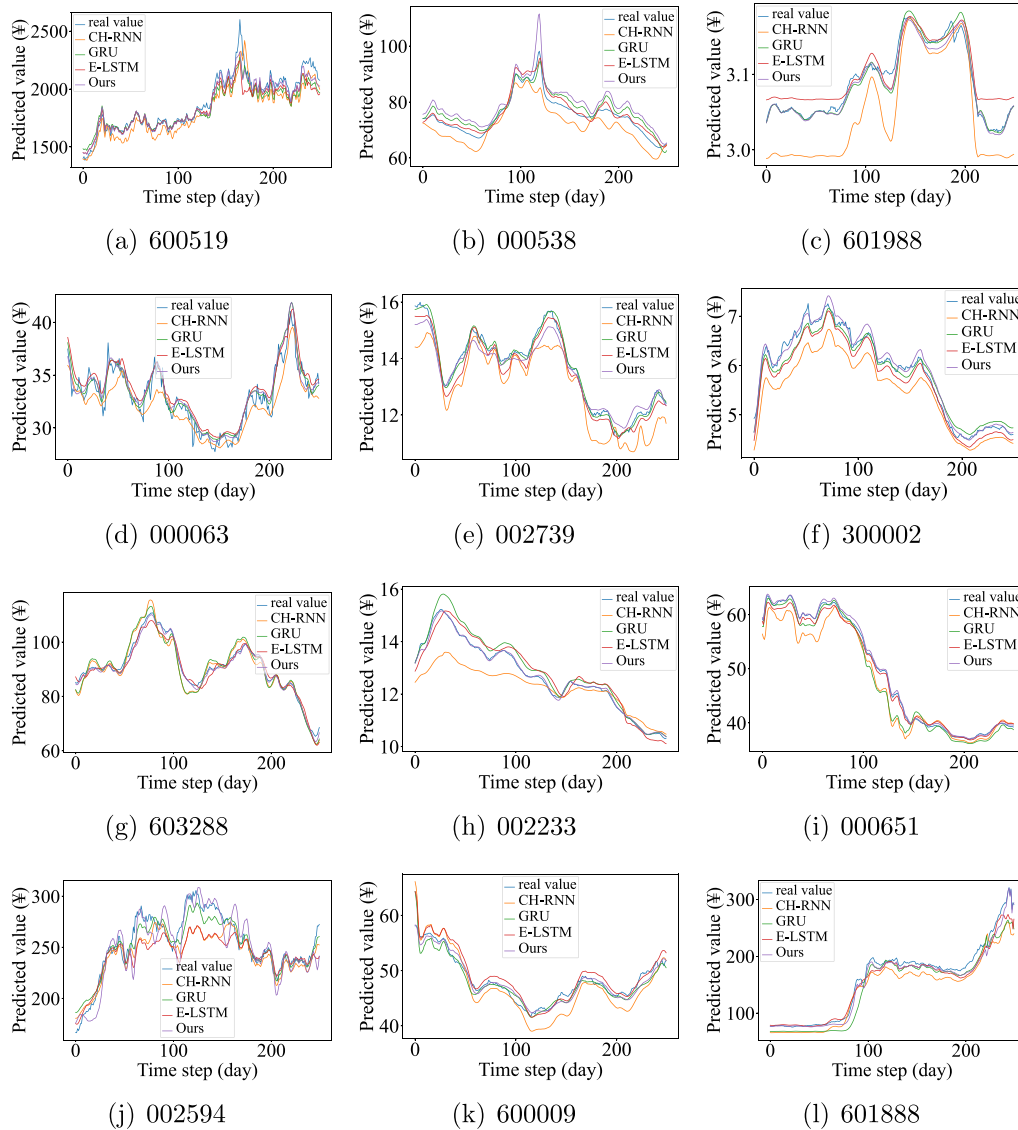


Fig. 7. Comparison of the different methods for all 12 stocks.

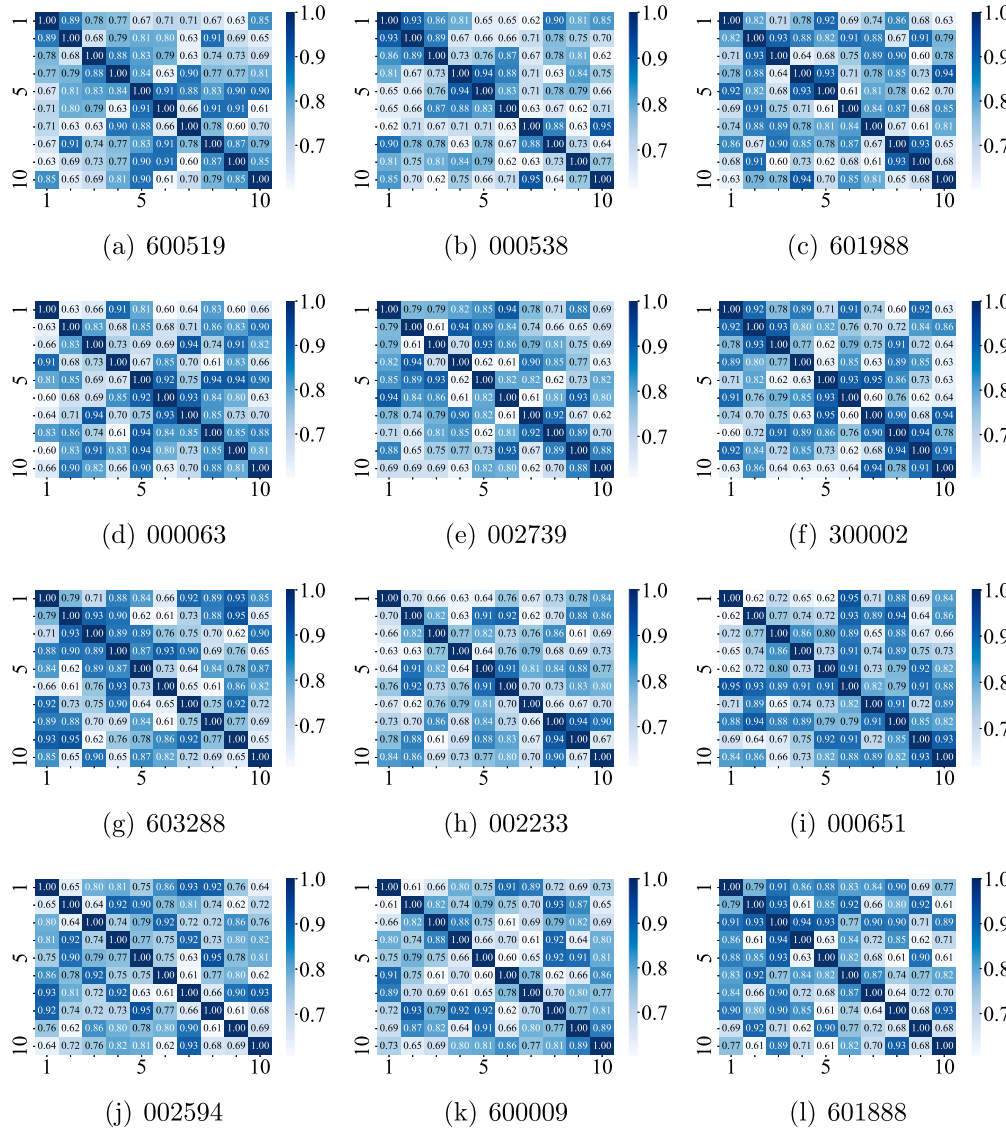


Fig. 8. PCC chart of all 12 stocks.

Table 7

CIR of different methods.

Methods	600519	000538	601988	000063	002739	300002
CH-RNN	0.63	0.67	0.51	0.62	0.59	0.73
GRU	0.84	0.78	0.55	0.51	0.79	0.77
E-LSTM	0.79	0.81	0.51	0.62	0.65	0.69
Ours	0.92	0.76	0.67	0.66	0.78	0.82

Methods	603288	002233	000651	002594	600009	601888
CH-RNN	0.63	0.56	0.63	0.56	0.54	0.61
GRU	0.65	0.61	0.67	0.67	0.62	0.67
E-LSTM	0.71	0.59	0.74	0.52	0.59	0.70
Ours	0.74	0.66	0.73	0.61	0.62	0.71

Table 8

Computational efficiency of different methods.

Methods	CH-RNN	GRU	E-LSTM	Ours
Params	36 864	54 082	128 945	318 800
Training (h)	1.5	1.8	3	4.8

5.4. Forex markets

To examine the applicability of our method to forex markets, we took four FX from <https://cn.investing.com/>, which are CHF/CNY, EUR/USD, USD/CNY and USD/HKD, from July 1, 2015 to July 1, 2023. The comparison of the results of the different methods and the generated scenarios with 95% confidence intervals are presented in Fig. 9. In addition, the MSE, MAPE, and CIR indicators are listed in Table 10.

As the results show, for the experimental results of four different foreign exchanges, our method is the most outstanding in terms of MSE, MAPE and CIR indicators, which indicates that our method is still applicable in the foreign exchange market.

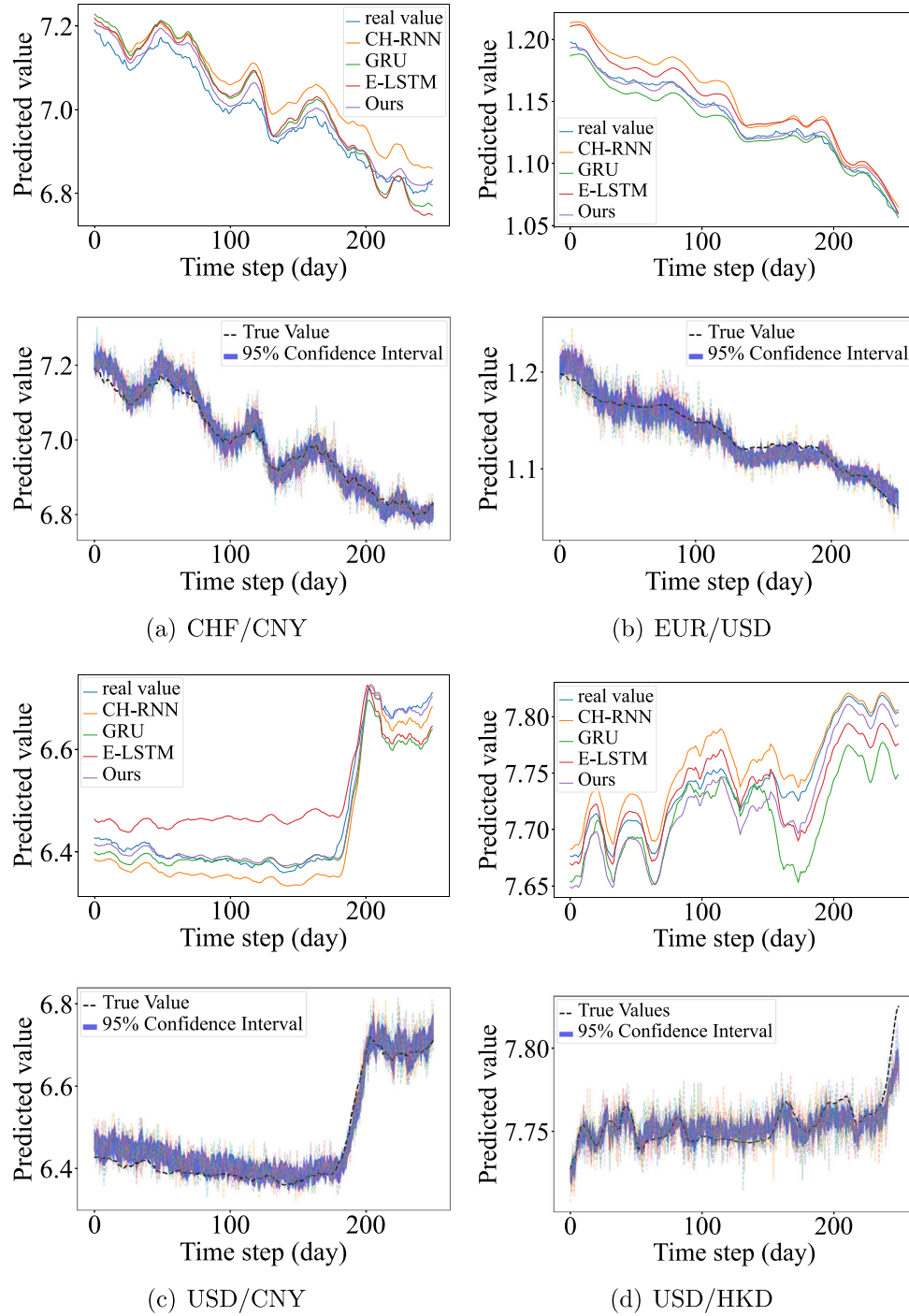
5.5. Incorporate expert knowledge

We also tried to collect predictions from experts on various social platforms and websites, labeling them with positive or negative tags based on the emotions expressed in the predictions. Then we introduced these tags into the WGAN to guide the results of scenario generation. Using this method, we conducted experiments on two stocks, presenting the results in Fig. 10 and Table 11.

Table 9

Scenario evaluation of ablation experiments (Wsp: without scenario generation. Wci: without cross-modal information. Wsc: without stock correlation).

Methods	600519		000538		601988		000063		002739	
	MSE	MAPE	MSE	MAPE	MSE	MAPE	MSE	MAPE	MSE	MAPE
Wsp	6923.2	4.25%	43.29	6.82%	0.0164	3.58%	5.68	6.04%	1.8671	6.38%
Wci	12171	4.63%	44.97	6.96%	0.0084	2.03%	5.44	5.62%	1.9643	6.44%
Wsc	55083	9.40%	51.67	8.24%	0.0126	7.39%	4.96	5.25%	1.6594	5.66%
Ours	5835.5	3.10%	45.22	7.35%	0.0057	1.55%	4.89	5.35%	1.8245	6.27%

**Fig. 9.** Experiment results of forex markets.

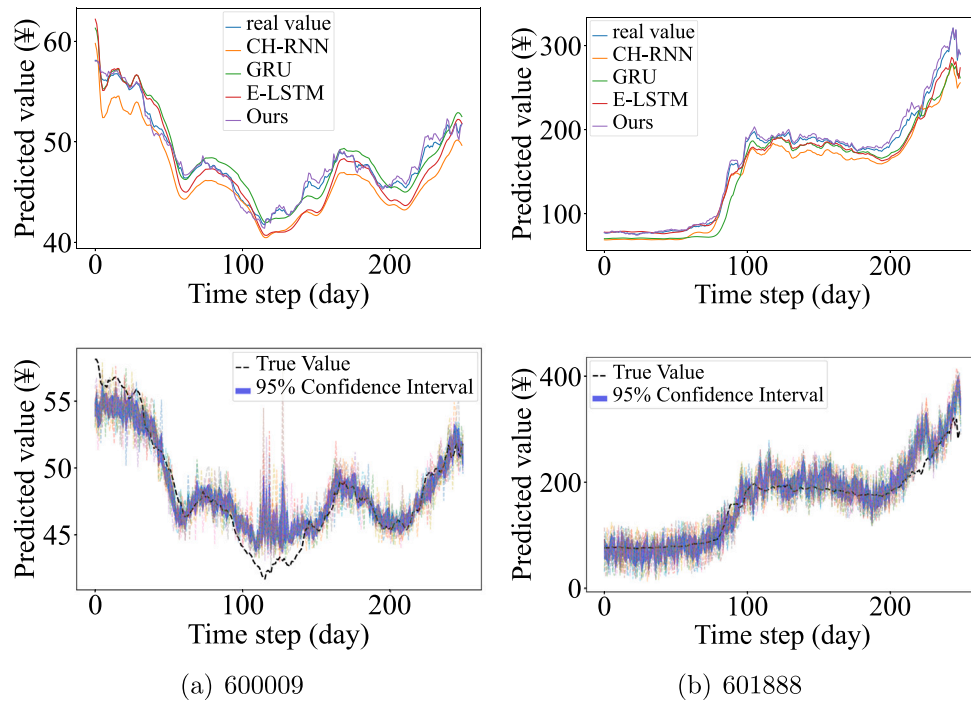


Fig. 10. Experiment results of incorporating expert knowledge.

Table 10

MSE, MAPE and CIR of different methods in forex markets.

Methods	CHF/CNY			EUR/USD		
	MSE	MAPE	CIR	MSE	MAPE	CIR
CH-RNN	0.0232	1.90%	0.66	8.65e-04	2.26%	0.51
GRU	0.0217	1.78%	0.73	7.99e-04	2.11%	0.57
E-LSTM	0.0218	1.78%	0.73	5.26e-04	1.72%	0.66
Ours	0.012	1.29%	0.85	3.99e-04	1.44%	0.76

Methods	USD/CNY			USD/HKD		
	MSE	MAPE	CIR	MSE	MAPE	CIR
CH-RNN	0.1694	5.90%	0.68	0.0299	1.75%	0.81
GRU	0.0817	3.91%	0.74	0.1505	4.71%	0.72
E-LSTM	1.7315	15.84%	0.66	0.0146	1.45%	0.85
Ours	0.006	0.91%	0.81	0.0022	0.47%	0.91

Table 11

MSE, MAPE and CIR of incorporating expert knowledge.

Methods	600009			601888		
	MSE	MAPE	CIR	MSE	MAPE	CIR
CH-RNN	72.9534	10.58%	0.59	2318.60	18.18%	0.65
GRU	41.5603	8.21%	0.64	1844.71	15.65%	0.70
E-LSTM	30.0324	7.41%	0.61	1474.89	14.22%	0.73
Ours	28.3035	7.32%	0.66	1237.47	13.15%	0.74

As the results show, following this method indeed leads to a certain improvement in effect. This time, we only made a simple attempt. In our future work, we will conduct more in-depth research.

5.6. Real market performance

In order to further justify the practical application effect of the proposed method, we adopted the portfolio selection model given in Section 4.5 and take 10,000 CNY as the total capital. Stocks listed in Table 4 are considered as the candidate investment targets, while point forecast as well as scenario generation were achieved by using our method and three baseline methods (CH-RNN, GRU, E-LSTM), i.e. totaling 8 groups of experiments. For each stock, 10 scenarios are

Table 12

Investment return of different methods.

Models	Point forecast		Scenario generation	
	Profits	Return(%)	Profits	Return(%)
CH-RNN	621.73	6.22%	748.34	7.48%
GRU	1039.19	10.39%	1109.37	11.09%
E-LSTM	1571.47	15.71%	1678.96	16.79%
Ours	1588.23	15.88%	1843.11	18.43%

generated while each scenario represents the future stock price over the next 30 days.

Generally, 30 days corresponds to 20 trading days. But it is not enough to consider such a short time frame. So we divided a total of 100 trading days into 5 stages. For each group of experiments, portfolio optimization was carried out on every 20 trading days using the portfolio model (35), and we set $R = 0.9$, $\alpha = 0.9$ and $\beta = -0.02$. After 100 days, the profit margins obtained by our method are compared with those of other methods, and the results are shown in Table 12.

To better visualize the effect of our approach, the real returns for each of the five stages are depicted in Fig. 11.

It can be seen that the simulated investment results of our method can obtain the highest return rate as 18.43%, which proves the effectiveness of CM-SG in practical application. At the same time, with the help of multiple scenarios, the investment return is greater than the point forecast, which also proves the core of this paper, i.e. scenario generation is conducive to make better investment decisions.

6. Conclusion and discussion

This paper realizes the scenario generation of stock price by using advanced machine learning technologies. Considering the correlation between stocks, we select a number of stocks with the largest PCC relative to the target stock, using GCN to learn the features of related stocks. Then we establish attention mechanism to fully learn as well as integrate the news text features and data features of these stocks. And we finally use WGAN to make scenario generation of stock prices.

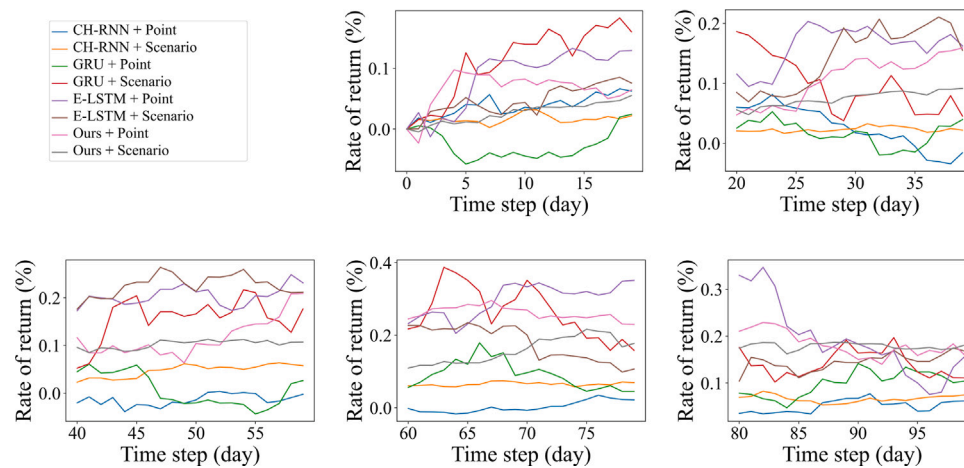


Fig. 11. Real market returns of different methods in five stages.

Experiments show that our method performs better than other baseline methods. Subsequent ablation experiments also show that the correlations, cross-modal feature fusion, and scenario generation considered in our approach can improve the accuracy of the final results. The result of our scenario generation can also be used as input to realize the solution of the portfolio model. The experiment of foreign exchange markets proves the generalization of our method. The simulation results show that our method can achieve 18.43% return within 100 trading days, which is higher than other methods, indicating that scenario generation can help investors comprehensively assess risks, optimize their investment strategies, and obtain higher returns.

In conclusion, our method can effectively increase the yield in the real world scenario, and can reduce some risks. However, our approach does have some limitations, such as difficulty adapting to highly volatile or extreme market conditions, which is a potential challenge. In future work, this study can be further improved in the future from the following aspects. Consider small sample events, such as large changes in stock prices caused by extreme events, to make our method suitable for highly volatile or extreme market conditions. Use state-of-the-art versions of Transformer to improve the accuracy of time series stock price forecasting. For the part of scenario generation, consider using the latest diffusion model to enhance the effect of the generated scenarios. In summary, future research will continue to focus on the development of stock price scenarios.

CRedit authorship contribution statement

Zixu Wang: Writing – original draft, Software, Methodology. **Bo Wang:** Writing – review & editing, Software, Resources. **You Li:** Software, Investigation. **Shu Liu:** Resources, Investigation. **Huaxiong Li:** Writing – review & editing, Visualization, Investigation. **Junzo Watada:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 72471109). The authors would like to express their sincere gratitude to the reviewers, whose valuable suggestions were conducive to the quality of this paper.

Data availability

Data will be made available on request.

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